

A Study of Calculus of Variations

Project in Calculus of Variations

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INTRODUCTION

In this project we will study and solve six calculus of variations problems with gradually increasing difficulty.

In the first chapter, Coaxial Bubble Rings, we will derive the curve that describes the shape of a bubble that is formed between two different sized rings coated with soap and then we will find the critical distance where this bubble pops. The focus then shifts to aerodynamics, where, in Chapter 2 - Formula 1 Venturi Vent, we analyze the Half-Car model in order to obtain the curve for the two-dimensional cross-section of the right part of the Venturi channel.

Next, we turn our attention to the domain of optics, specifically examining the initial conditions necessary to initiate meridional and skew rays as well as their trajectory inside a GRIN (Gradient-Index) waveguide (Chapter 3 – GRIN Meridional and Skew Rays). Following this, we shift to the analysis of some classical mechanics problems. More precisely, we will study the trajectory of a spring pendulum as well as the trajectory of a mass moving inside a paraboloid (Chapters 4 – Spring Pendulum and 5 – Mass Movement Inside a Paraboloid).

Finally, the project culminates in an analysis of general relativity where, using the Schwarzschild metric, we will find the initial conditions and the trajectory that a spacecraft needs to follow in order to land in a specific location, minimizing the time that has passed on Earth.

All the problems that we described above are going to be solved mathematically and then they will be coded and simulated with MATLAB (versions 2019a and 2024a).

COAXIAL BUBBLE RINGS

Problem Description and Objectives

In this problem two coaxial rings, with radii $R_1 = 1, R_2 = 0.8$ respectively, with negligible thickness are submerged next to each other in a soap water solution. As we move them apart from each other along the direction of their axis (x-axis) a very thin soap film is formed, symmetric with respect to the same axis. The center of the first ring ends up at the point $x = 0$, while the center of the second ring ends up at the point $x = L = 1$. The surface tension forces tend to minimize the area of the formed film.

The main objectives of this particular problem are:

1. To find the curve $y(x)$ which, after rotation around the x-axis, describes the shape of the film.
2. To find the physical limit, L_{crit} , of how far apart the rings can be moved from each other where if we surpass that limit the curved surface bubble will break

instantly. We know that this happens if the surface area of the curve exceeds the total area of the two flat disks.

Solution

The first thing we need to do in order to solve this problem is to derive the surface functional that we need to integrate and then minimize it with the Euler Lagrange (EL) method.

We can find the functional, $f(x, y, y')$, relatively simply if we think of it as a normal curve between 2 points multiplied by the circumference of a circle, which gives us:

$$dS = 2\pi y \sqrt{1 + (y')^2} dx$$

Which means that the surface we need to minimize is:

$$S = \int_0^L 2\pi y \sqrt{1 + (y')^2} dx \quad (1.0)$$

And since $L = 1$:

$$S = \int_0^1 2\pi y \sqrt{1 + (y')^2} dx$$

So, our functional is:

$$f(y, y') = 2\pi y \sqrt{1 + (y')^2} \quad (1.1)$$

Now if we use the EL formula, we get that:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0$$

Where, by simple differentiation of (1) we get:

$$\frac{\partial f}{\partial y} = 2\pi y \sqrt{1 + (y')^2} \quad (1.2)$$

$$\frac{\partial f}{\partial y'} = 2\pi y \frac{y'}{\sqrt{1 + (y')^2}} \quad (1.3)$$

Now, if we combine (1.1), (1.2) and (1.3) we arrive at:

$$\sqrt{1 + (y')^2} - \frac{d}{dx} \left(y \frac{y'}{\sqrt{1 + (y')^2}} \right) = 0$$

which after some refinements becomes:

$$y'' = \frac{1 + (y')^2}{y} \quad (1.4)$$

In order to code the solution with MATLAB we need to rewrite (1.4) as a first-order system which yields:

$$\begin{cases} y_1 = y \\ y_2 = y' \end{cases} \Rightarrow \begin{cases} y_1' = y_2 \\ y_2' = y'' = \frac{1 + (y')^2}{y} \end{cases} \quad (1.5)$$

Also, since we know the 2 radiuses we can derive the boundary conditions of the problem, which will be:

$$\begin{cases} R_1 = 1 \\ R_2 = 0.8 \end{cases} \Rightarrow \begin{cases} y(0) = 1 \\ y(1) = 0.8 \end{cases}$$

The code that will solve this system of equations, and our problem, is presented below:

Table 1 – BubbleRing.m Part A

```
%% A) Find y(x)
clc;
clear;

% Initialize the parameters
R1 = 1;
R2 = 0.8;

L = 1; % Distance between the two rings
S = pi*R1^2 + pi*R2^2; % Surface of the 2 rings combined
step = 0.1; % Step that increases the distance between the rings

xmesh = linspace(0, L, 100);
solinit = bvpinit(xmesh, @guess);

% Solve the boundary value problem
sol = bvp4c(@odefun, @bcfun, solinit);

% Plot the solution
x = sol.x;
y = sol.y(1, :);
yp = sol.y(2, :);

figure;
plot(x, y, 'b-'); grid;
hold on;
plot(x, -y, 'b-');

plot([0 0], [R1 -R1], 'k-');
plot([1 1], [R2 -R2], 'k-');
title('2D Line Solution of the min y(x)')

% --- 3D plot ---
theta = linspace(0, 2*pi, 100);

[THETA, x_3D] = meshgrid(theta, x);
```

```

y_radius = repmat(y', 1, length(theta));

y_3D = y_radius .* cos(THETA);
z_3D = y_radius .* sin(THETA);

figure;
surf(x_3D, y_3D, z_3D, 'EdgeColor', 'none', 'FaceAlpha', 0.6); title('3D Solution Bubble');
hold on;

% Plot the rings
plot3(zeros(1,100), R1*cos(theta), R1*sin(theta), 'k-', 'LineWidth', 3);
plot3(L*ones(1,100), R2*cos(theta), R2*sin(theta), 'k-', 'LineWidth', 3); grid on;

% --- Analytic Solution ---

syms y(x)
DE = diff(y,x,2) == ( 1 + diff(y,x)^2 ) / y;

sol = dsolve(DE);

disp('The analytic equation is y(x):');
disp(sol);

function dydx = odefun(x, y)
    dydx = zeros(2,1);
    dydx(1) = y(2); % y1' = y2
    dydx(2) = (1 + y(2)^2) / y(1); % y2' = (1 + y2^2) / y1
end

function res = bcfun(ya, yb)
    res = [ ya(1) - 1; % y(0) = 1
            yb(1) - 0.8]; % y(1) = 0.8
end

function yinit = guess(x)
    yinit = [1; 0];
end

```

This code above initializes all the problem parameters, the boundary conditions and the first-order system of equations and uses the MATLAB function `bvp4c()` in order to solve this boundary value problem. It then, for output, it gives us the analytic solution of $y(x)$ and plots 2 figures. The first figure is a 2D plot of the minimal line solution $y(x)$ and the second figure is 3D plot of the surface of the bubble.

Our outputs are presented below:

```

The analytic equation is y(x):

C2 - x*1i
C3 + x*1i
(exp(C1^(1/2)*(C4 + x)) + exp(-C1^(1/2)*(C4 + x)))/(2*C1^(1/2))
(exp(C1^(1/2)*(C5 - x)) + exp(-C1^(1/2)*(C5 - x)))/(2*C1^(1/2))

```

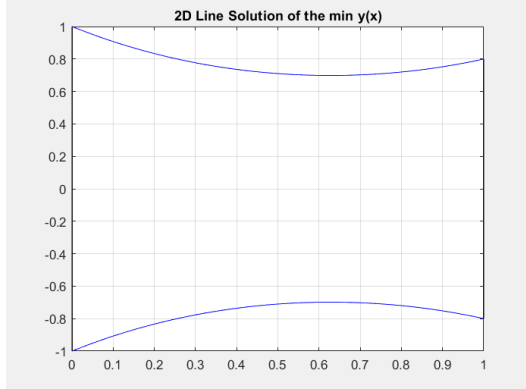


Figure 1. 2D Line Solution of the min $y(x)$

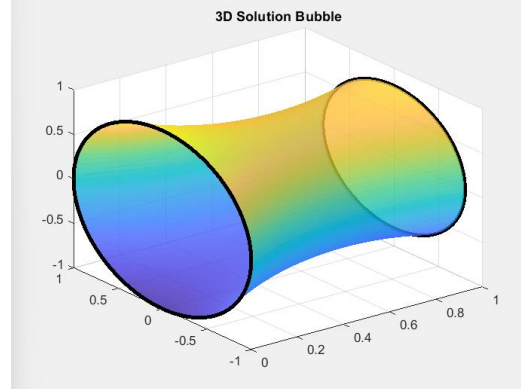


Figure 2. 3D Solution Bubble

Now the only thing left is to find the critical limit between the 2 rings, L_{crit} . In order to achieve this, we need to slightly modify the code given above so that it can calculate the integral of the area, S , for different values of L . To do this, we will set a starting separation length value at $L=0.1$, increment it by a step ($step=0.01$, when tested) inside a loop, and check each time if the value of S , (1.0), (solved with MATLABs `trapez()` function) is larger than the total area of the two flat disks. Again, the solution for every different L value will be calculated via the function `bvp4c()`. The code that does this is the following:

Table 2. BubbleRing.m Part B

```
%% B) Which is the Lcrit where the bubble bursts?
clear;

% Initialize the radiuses
R1 = 1;
R2 = 0.8;

L = 0.1; % Distance between the two rings
S = pi*R1^2 + pi*R2^2; % Surface of the 2 rings combined
step = 0.01; % Step that increases the distance between the rings

xmesh = linspace(0, L, 100);
solinit = bvpinit(xmesh, @guess);

% Solve the boundary value problem
sol = bvp4c(@odefun, @bcfun, solinit);

% Plot the solution
x = sol.x;
y = sol.y(1, :);
yp = sol.y(2, :);

integrand = y .* sqrt(1 + yp.^2);
surf_3D = 2*pi * trapz(x, integrand);

while true
    % Break if the area if surf_3D is larger than S
    if surf_3D >= S
        disp(['The bubble will burst since surf_3D = ', num2str(surf_3D), ' > S = ',
num2str(S)]);
        disp(['Lcrit = ', num2str(L)]);
        break;
    end
    L = L + step;
    S = pi*R1^2 + pi*R2^2;
    xmesh = linspace(0, L, 100);
    solinit = bvpinit(xmesh, @guess);
    sol = bvp4c(@odefun, @bcfun, solinit);
    y = sol.y(1, :);
    yp = sol.y(2, :);
    integrand = y .* sqrt(1 + yp.^2);
    surf_3D = 2*pi * trapz(x, integrand);
end
```

```

end

L = L + step;

xmesh = linspace(0, L, 100);
solinit = bvpinit(xmesh, @guess);

% Solve the boundary value problem
sol = bvp4c(@odefun, @bcfun, solinit);

x = sol.x;
y = sol.y(1, :);
yp = sol.y(2, :);

% Compute the new area
integrand = y .* sqrt(1 + yp.^2);
surf_3D = 2*pi * trapz(x, integrand);
end

function dydx = odefun(x, y)
    dydx = zeros(2,1);
    dydx(1) = y(2); % y1' = y2
    dydx(2) = (1 + y(2)^2) / y(1); % y2' = (1 + y2^2) / y1
end

function res = bcfun(ya, yb)
    res = [ ya(1) - 1; % y(0) = 1
            yb(1) - 0.8]; % y(1) = 0.8
end

function yinit = guess(x)
    yinit = [1; 0];
end

```

The output of the code above gives us:

```

The bubble will burst since surf_3D = 5.1548 > S = 5.1522
Lcrit = 0.94

```

Which means that the critical value of L is: $L_{crit} = 0.94m$

FORMULA 1 VENTURI CHANNEL

Problem Description and Objectives

Here we undertake the optimal design of the aerodynamic floor of a Formula 1 single-seater, and specifically of the Venturi Channel. Due to the absolute geometric and aerodynamic symmetry of the vehicle, our study will be based on the analytical "Half-Car Model". Therefore, we are going to design the two-dimensional cross-section of the right part of the channel which is defined by the curve $y(x)$ and extends from the central longitudinal axis of symmetry of the single-seater, where $x = 0$, to the outer edge of the floor, where $x = w = 1 \text{ m}$, with $y(w) = h_{seal} = 0.06 \text{ m}$.

The mechanical purpose of this geometry is twofold: On one hand, the channels must be occupied by a certain volume of air at all times, causing a drop in static pressure (based on Bernoulli's principle), in order to produce the required vertical aerodynamic force (Downforce) that ensures a part of the aerodynamic grip of the single-seater. On the other hand, the cross-sectional profile must be designed in such a way as to minimize the resulting drag, which develops due to the friction of the air with the walls of the floor.

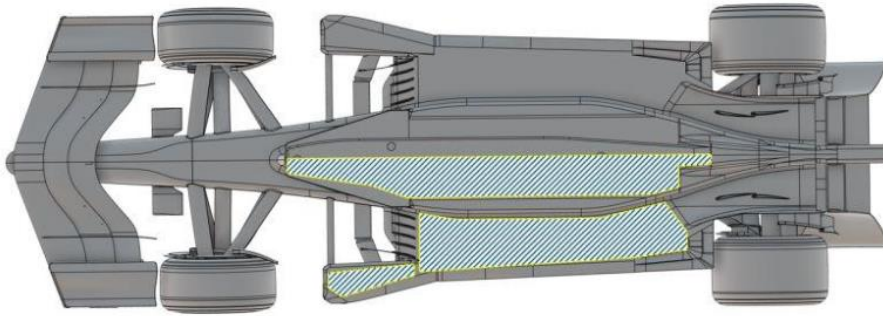


Figure 3. Formula 1 Venturi Channel

The objectives of this chapter are to:

1. Design the curve $y(x)$, which will have the minimum possible length, given that the cross-sectional area must be $A_0 = 0.15 \text{ m}^2$.
2. Test 3 different values for A_0 and explain the physical constraints and the effects of the area variation on the car's behavior.

Solution

To solve this problem, we must realize that we are talking about a constrained variational problem, since we need to minimize the arc length under a specific area A_0 , which means that we need to use Lagrange's method (the Lagrange multiplier λ). The arc length can be described, as we did in the previous chapter, by the following integral:

$$I(y) = \int_0^w \sqrt{1 + (y')^2} dx = \int_0^1 \sqrt{1 + (y')^2} dx \quad (2.1)$$

While the limiting area, A_0 , under the arc:

$$J(y) = \int_0^1 y dx = A_0 = 0.15m^2 \quad (2.2)$$

So, from Lagrange's solution for constrained variational problems we tie the two integrals, (2.1) and (2.2), and the integral we need to minimize becomes:

$$H(y) = I(y) + \lambda \cdot J(y) = \int_0^1 \sqrt{1 + (y')^2} + \lambda y dx \quad (2.3)$$

Which means that our functional is:

$$h(x, y, y') = \sqrt{1 + (y')^2} + \lambda y \quad (2.4)$$

where λ is the Lagrange multiplier.

If we now use the EL formula:

$$\frac{\partial h}{\partial y} - \frac{d}{dx} \frac{\partial h}{\partial y'} = 0$$

We get that:

$$\frac{\partial h}{\partial y} = \lambda \quad (2.5)$$

$$\frac{\partial h}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2}} \quad (2.6)$$

Now, if we combine (2.4), (2.5) and (2.6) and substitute them in the EL formula we get:

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2}} \right) = \lambda \quad (2.7)$$

If we integrate (2.7) we arrive at:

$$\frac{y'}{\sqrt{1 + (y')^2}} = \lambda x - C_1 \quad (2.8)$$

Now, if we square both sides of (2.8) and after some reforming we get:

$$dy = \frac{\lambda x - C_1}{\sqrt{1 - (\lambda x - C_1)^2}} dx \quad (2.9)$$

which after integration becomes:

$$y(x) = \frac{-1}{\lambda} \sqrt{1 - (\lambda x - C_1)^2} + C_2 \quad (2.10)$$

From the natural boundary condition, we have that:

$$\left. \frac{\partial h}{\partial y'} \right|_{x=0} = 0 \Rightarrow y'(0) = 0 \quad (2.11)$$

If we substitute (2.11) in (2.9) we get that:

$$C_1 = 0 \quad (2.12)$$

Now since $y(1) = 0.06 \text{ m}$, from (2.10) and (2.12) we get:

$$y(1) = \frac{-1}{\lambda} \sqrt{1 - \lambda^2} + C_2 = 0.06 \quad (2.13)$$

We also have (2.2) which if we combine it with (2.10), (2.12) and calculate the integral with MATLAB we arrive at:

$$C_2 - \frac{\arcsin(\lambda)}{2\lambda^2} - \frac{\sqrt{1 - \lambda^2}}{2\lambda} = 0.15 \quad (2.14)$$

Finally, if we can use the system of (2.13) and (2.14) to find our solution curve.

The code that does all of the above in MATLAB is presented below:

Table 3. F1_VenturiVent.m

```
clc;
clear;

% Constants
A0 = 0.15; % A0 = x m^2 is the cross-section area under y(x)

% Initial guess
x0 = [-0.1; -0.1];

x = fsolve(@(x) F(x, A0), x0);

% Display results
disp(['Lambda: ' num2str(x(1))])
disp(['c_2: ' num2str(x(2))])
```

```

% Check results
disp(F(real(x), A0));

% Create plot
x = real(x);
t = 0:0.001:1;
y = (-1/x(1)) * sqrt( 1 - (x(1)*t).^2 ) + x(2);
plot(t,y, 'LineWidth', 3);
xlabel('x');
ylabel('y');
title(['Optimal Arc for Venturi Vent with Cross-Sectional Area A0: ', num2str(A0), 'm^2']);
hold on;
grid on;
axis equal;
ylim([0, inf]);

% plot([0, 1], [0.06, 0.06]);
hold off;

% x1 = lamda, x2 = c2
function y = F(x, A0)
    y(1) = ( -1/x(1) ) * sqrt( 1 - x(1)^2 ) + x(2) - 0.06;
    y(2) = x(2) - (asin(x(1))/2 + (x(1)*(1 - x(1)^2)^(1/2))/2)/x(1)^2 - A0;
end

```

The code above initializes the wanted cross-sectional area A_0 and then passes the equations (2.13) and (2.14) into MATLAB's `fsolve()` which then gives us the solution curve we want. Then it proceeds to plot the arc, as shown below.

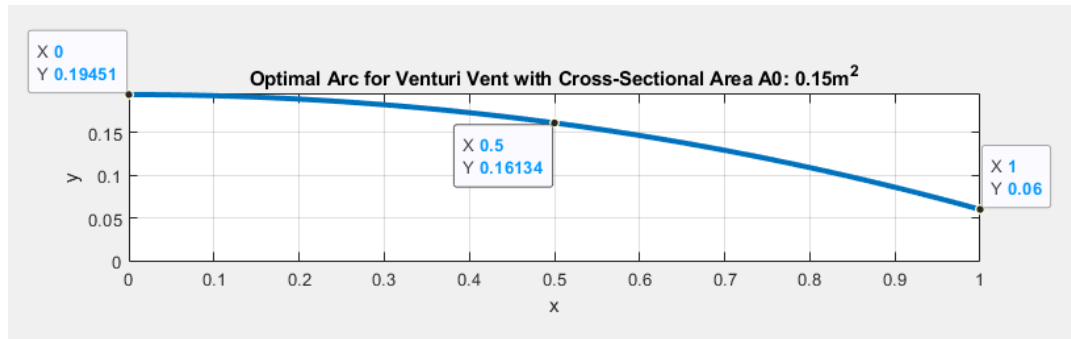


Figure 4. Optimal Arc for Venturi Vent with area $A_0 = 0.15 \text{ m}^2$

Now, we are going to try 3 different values for the cross-sectional area A_0 under our curve and then find and plot the solutions in order to try to explain the physical constraints and the effects of the area variation on the car's behavior.

The values of A_0 we are going to try are: $A_0 = 0.1 \text{ m}^2$, 0.3 m^2 and 0.6 m^2 . Their solution plots and our conclusions are presented below:

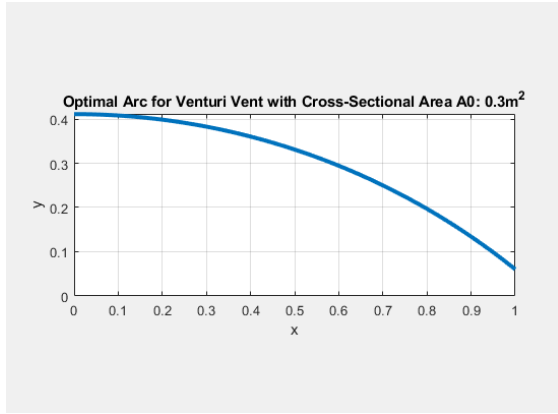


Figure 5. Optimal Arc for Venturi Vent with area $A_0 = 0.3 \text{ m}^2$

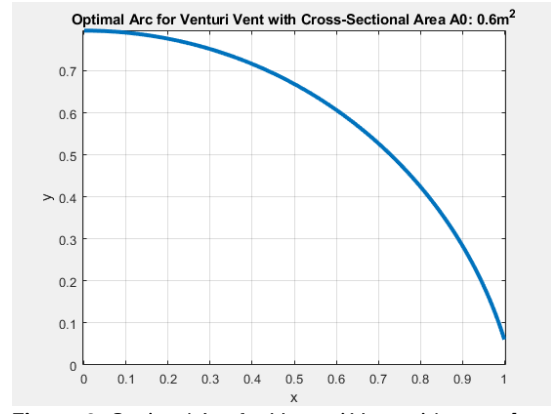


Figure 6. Optimal Arc for Venturi Vent with area $A_0 = 0.6 \text{ m}^2$

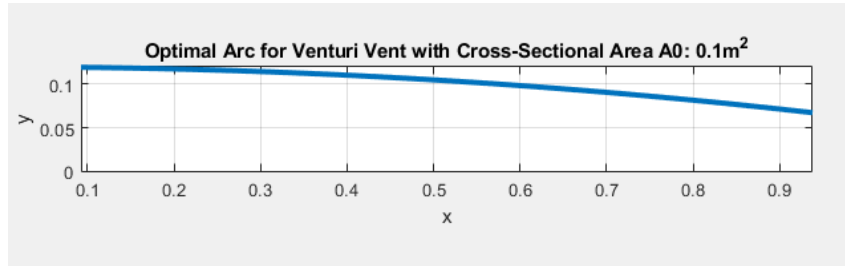


Figure 7. Optimal Arc for Venturi Vent with area $A_0 = 0.1 \text{ m}^2$

Attempting to design an excessively deep Venturi Tunnel by drastically increasing A_0 forces the height of the curve at the central axis of symmetry to rise significantly. From a vehicle packaging perspective, this mathematical ideal would require the floor's roof to intrude into spatial volumes already occupied by critical components, such as the driver's seat, the engine, and the battery store. Aerodynamically, the airflow possesses inertia and cannot remain attached to the aggressive curvature of an oversized arc. This leads to boundary layer separation, the creation of turbulent vortices, and a subsequent collapse of downforce, known as aerodynamic stall. Furthermore, according to the continuity equation and Bernoulli's principle, a massive cross-sectional area drastically reduces the airflow velocity, which inadvertently increases static pressure and neutralizes the desired ground effect. Finally, the increased arc length required for a larger area maximizes the wetted surface of the carbon fiber, resulting in a severe penalty in parasitic skin friction drag.

Conversely, designing an extremely shallow channel with a minimized A_0 forces the floor profile dangerously close to the track surface. At high speeds, the aerodynamic load compresses the suspension, risking violent contact between the floor and the asphalt, which can cause severe wear and a sudden loss of vehicle control. Moreover, this minimal spatial gap causes the boundary layer of the floor to merge with the boundary layer of the track. This phenomenon, known as choked flow, physically blocks the air from passing under the car. The abrupt alternation between the generation of extreme suction and the immediate choking of the flow triggers violent vertical oscillations, widely recognized as porpoising, rendering the car highly unstable on long straights.

GRIN MERIDIONAL AND SKEW RAYS

Problem Description and Objectives

Consider a cylindrical plasma chamber of radius $R = 0.1 \text{ m}$ configured as a zero-loss gradient-index (GRIN) waveguide for laser acceleration experiments

The length of the chamber is $L=1 \text{ m}$ and the refractive index of the plasma is $n(r) = n_0 \exp(-\alpha r^2)$, where $n_0 = 1$ on the central z -axis, and $\alpha = 50$ is the focusing parameter. The initial conditions of the problem relate to the entry point and entry angle of the laser rays.

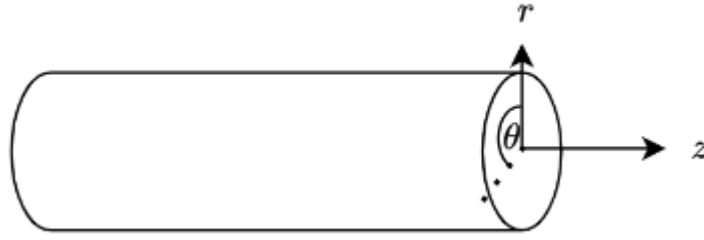


Figure 8. GRIN waveguide

The objectives of this chapter are to:

1. Define the initial conditions and explain their physical meaning, so that the ray behaves as a Skew Ray and as a Meridional Ray.
2. In reality, the refractive index of the plasma depends on the wavelength of the ray. Assuming this time that $\alpha(\lambda) = 20 \left(1 + \left(\frac{\lambda_0}{\lambda} \right)^2 \right)$, where $\lambda_0 = 600 \text{ nm}$, examine the behavior of the ray inside the waveguide, if it consists of a red HeNe ray, with $\lambda_r = 633 \text{ nm}$, and a green Argon ray, with $\lambda_g = 514 \text{ nm}$, and comment on the results.

Meridional Ray Solution

Since we know by definition that meridional rays propagate in a 2D-plane we only need to solve the 2D problem here. Also, we are going to use the cylindrical coordinate system (r, θ, z) .

In order to solve this problem, we are going to use Fermat's Principal. So, we have that the two-dimensional speed of light is given by:

$$c(r) = \frac{ds}{dt} \quad (3.1)$$

The time required to cover the distance between two points is:

$$t = \int \frac{1}{c(r)} ds \quad (3.2)$$

The problem is now posed as a variational problem of:

$$I(r) = \int_{(z_1, r_1)}^{(z_2, r_2)} \frac{1}{c(r)} ds \quad (3.3)$$

Where:

$$ds = \sqrt{dr^2 + dz^2} \quad (3.4)$$

Then, since we know that:

$$c(r) = \frac{c_0}{n(r)} = \frac{c_0}{n_0 \exp(-ar^2)}$$

and that: $\alpha = 50, n_0 = 1$, by combining $c(r)$ with (3.3) and (3.4) we arrive at:

$$S = \int_{z_1}^{z_2} \frac{e^{-50r^2}}{c_0} \sqrt{1 + (r')^2} dz \quad (3.5)$$

From here we can derive our functional:

$$f(r, r') = \frac{e^{-50r^2}}{c_0} \sqrt{1 + (r')^2} \quad (3.6)$$

which enables us to calculate:

$$\frac{\partial f}{\partial r} = -100e^{-50r^2} \frac{r}{c_0} \sqrt{1 + (r')^2} \quad (3.7)$$

$$\frac{\partial f}{\partial r'} = \frac{e^{-50r^2}}{c_0} \frac{r'}{\sqrt{1 + (r')^2}} \quad (3.8)$$

Now, we can substitute (3.7) and (3.8) into the EL formula and get:

$$-100e^{-50r^2} \frac{r}{c_0} \sqrt{1 + (r')^2} - \frac{d}{dz} \left(\frac{e^{-50r^2}}{c_0} \frac{r'}{\sqrt{1 + (r')^2}} \right) = 0$$

which after differentiating with MATLAB and rearranging we get:

$$r'' = -100r(1 + (r')^2) \quad (3.9)$$

Then, we rewrite (3.9) into a system of two first-order equations as shown below:

$$\begin{cases} r_1 = r \\ r_2 = r' \end{cases} \Rightarrow \begin{cases} r_1' = r_2 \\ r_2' = r'' = -100r(1 + (r')^2) \end{cases} \quad (3.10)$$

so that we can pass them in MATLABs function ode45().

Also, in order for the ray to propagate as a meridional ray we need to define the initial conditions. Meridional rays propagate into 2D planes, so we only need to initialize 2 values, the starting position at the r-axis, $r(0) = r_0 < 0.1$, and the shooting angle, $r'(0) = \frac{dr}{dz}\bigg|_{z=0}$.

The code that does all of the above is presented below:

Table 4. GRIN_Meridional_Ray.m

```

clc;
clear;
tic
% --- Initial Conditions & Z Limit ---
% y0 = [r(0), r'(0)];
y0 = [0.05; 0.5];
L = 1;
zspan = [0, L];
% a = 50;
lamda0 = 600; % 600nm
lamda_r = 633; % 633nm for HeNe
lamda_g = 514; % 514nm for Argon
a = [20*(1 + (lamda0/lamda_r)^2), 20*(1 + (lamda0/lamda_g)^2)];
color = ['r', 'g'];

figure;
hold on;
for i = 1:2

    % --- Set up the 2 ODEs ---
    % Set y1 = r, y2 = r'
    odesys = @(z, y) [
        y(2); % y1' = r' = y2
        -2*a(i)*y(1)*(1+y(2)^2); % y2' = r''
    ];

    % --- Solve with ode45 ---
    [z_sol, y_sol] = ode45(odesys, zspan, y0);

    % --- 3D Plot ---
    X_val = y_sol(:,1);
    Y_val = zeros(size(z_sol));

    plot3(z_sol, X_val, Y_val, color(i), 'LineWidth', 1.5);
end

[Xc, Yc, Zc] = cylinder(0.1, 50);
Zc = Zc * L;
surf(Zc, Xc, Yc, 'FaceAlpha', 0.1, 'EdgeColor', 'none', 'HandleVisibility', 'off',
'FaceColor', 'c');

grid on;
xlabel('Z (m) - Cylindrical Waveguide Length');
ylabel('X (m) - Radius r');
zlabel('Y (m)');
title(['Meridional Rays in GRIN Waveguide with a_r: ', num2str(a(1)), ' and a_g',
num2str(a(2))]);
legend('HeNe (\lambda_r = 633 nm)', 'Argon (\lambda_g = 514 nm)', 'Location', 'best');

```



```

axis equal;
zlim([-0.1 0.1]);
ylim([-0.1 0.1]);
view(3);

disp('Done!');
toc

```

This code outputs the plot of the propagation of both the red, HeNe, and the green, Argon, rays simultaneously. The initial conditions for this result were $[r(0); r'(0)] = [0.05; 0.5]$.

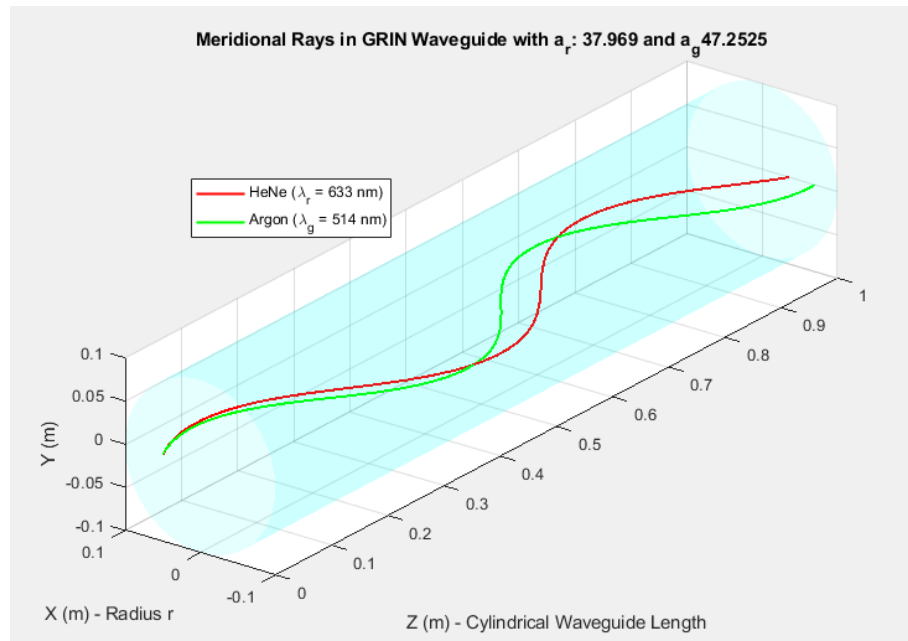


Figure 9. Meridional Rays in a GRIN Waveguide

Skew Ray Solution

SPRING PENDULUM

MASS MOVEMENT INSIDE A PARABOLOID

SCHWARZSCHILD